Chapter 2 — Atomic Structure

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

Practice MCQs

1. Atomic number is the number of:

(a) Nucleons **(b)** Protons **(c)** Neutrons **(d)** Electrons in ion

2. Mass number equals:

(a) Protons + neutrons **(b)** Electrons only **(c)** Neutrons only **(d)** Protons + electrons

3. Neutrons in 27Al with Z=13:

(a) 14 **(b)** 27 **(c)** 40 **(d)** 13

4. Electrons in Al_3+ with Z=13:

(a) 10 **(b)** 16 **(c)** 13 **(d)** 27

5. In an electric field, electrons move toward:

(a) Negative plate **(b)** Positive plate **(c)** Both plates **(d)** No deflection

6. Neutrons in an electric field are:

(a) Deflected to + **(b)** Undeflected **(c)** Split **(d)** Deflected to -

7. Electrons deflect more than protons because:

(a) No charge **(b)** Larger size **(c)** Larger charge **(d)** Much smaller mass

8. Large jump in successive IE indicates:

(a) New bond **(b)** New neutron **(c)** New inner shell **(d)** New isotope

9. Principal quantum number gives:

(a) Orbital spin **(b)** Oxidation state **(c)** Bond angle **(d)** Shell size/energy

10. Azimuthal quantum number gives:

(a) Subshell shape **(b)** Shell number **(c)** Spin **(d)** Charge

11. For p subshell, l is:

(a) 0 **(b)** 1 **(c)** 3 **(d)** 2

12. Number of orbitals in p subshell:

(a) 1 **(b)** 5 **(c)** 7 **(d)** 3

13. Number of orbitals in d subshell:

(a) 5 **(b)** 1 **(c)** 7 **(d)** 3

14. Maximum electrons in d subshell:

(a) 10 **(b)** 2 **(c)** 6 **(d)** 14

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15. Spin quantum number values are:

(a) 1 and 2 **(b)** +1/2 and -1/2 **(c)** +1 and -1 **(d)** 0 and 1

16. Pauli exclusion principle says:

(a) Single fill first **(b)** Orbitals are circular **(c)** Fill lowest first **(d)** No same four quantum numbers

17. Aufbau principle says:

(a) Fill highest first **(b)** Pair first **(c)** Fill lowest energy first **(d)** Random filling

18. Hund rule says:

(a) Pair before single **(b)** Fill d before s **(c)** Skip orbitals **(d)** Singly fill degenerate orbitals

19. Shape of s orbital is:

(a) Spherical **(b)** Linear **(c)** Dumb-bell **(d)** Clover-leaf

20. Shape of p orbital is:

(a) Spherical **(b)** Ring **(c)** Dumb-bell **(d)** Octahedral

21. Number of d orbitals:

(a) 1 **(b)** 7 **(c)** 5 **(d)** 3

22. Ground-state configuration of N:

(a) 1s2 2s1 2p4 **(b)** 1s2 2p5 **(c)** 1s2 2s2 2p3 **(d)** 1s2 2s2 2p4

23. Configuration of Cr is exceptional due to:

(a) Half-filled d stability **(b)** No d electrons **(c)** Large size **(d)** Full p subshell

24. 4s fills before 3d because:

(a) Lower energy **(b)** Higher n always **(c)** Larger shape **(d)** More orbitals

25. Hydrogen line spectrum proves:

(a) No nucleus **(b)** No electrons **(c)** Quantized energy levels **(d)** Continuous energy

26. Balmer series lies in:

(a) X-ray **(b)** Visible region **(c)** IR **(d)** UV

27. Lyman series lies in:

(a) UV **(b)** Visible **(c)** Radio **(d)** IR

28. Atomic spectra are used to:

(a) Measure mass only **(b)** Identify elements **(c)** Find neutrons only **(d)** Make bonds

29. An orbital can hold at most:

(a) 2 electrons **(b)** 6 electrons **(c)** 8 electrons **(d)** 1 electron

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30. For n=3, number of subshells:

(a) 4 **(b)** 1 **(c)** 2 **(d)** 3

31. Proton number is also called:

(a) Neutron number **(b)** Atomic number **(c)** Mass number **(d)** Nucleon number

32. Nucleon number equals:

(a) Protons + electrons **(b)** Neutrons only **(c)** Electrons only **(d)** Protons + neutrons

33. In an electric field, protons move to:

(a) No deflection **(b)** Negative plate **(c)** Positive plate **(d)** Both plates

34. Neutrons in an electric field are:

(a) Deflected to positive plate **(b)** Split **(c)** Undeflected **(d)** Deflected to negative plate

35. Atomic spectra are:

(a) Line spectra **(b)** Absorption only **(c)** No spectra **(d)** Continuous spectra

36. Line spectra show that energy levels are:

(a) Absent **(b)** Continuous **(c)** Random **(d)** Quantized

37. A large jump in successive IE indicates:

(a) New inner shell **(b)** Same shell **(c)** Isotopes **(d)** No electrons

38. Azimuthal quantum number gives:

(a) Orientation **(b)** Spin **(c)** Shell size **(d)** Subshell shape

39. Magnetic quantum number gives:

(a) Spin **(b)** Subshell shape **(c)** Shell size **(d)** Orbital orientation

40. Spin quantum number has values:

(a) +1/2 and -1/2 **(b)** 0 only **(c)** +1 and -1 **(d)** 0 and 1

41. Pauli exclusion principle says no two electrons have same:

(a) Mass **(b)** Four quantum numbers **(c)** Radius **(d)** Charge

42. Aufbau principle says electrons fill:

(a) Lower energy first **(b)** Higher energy first **(c)** Only s orbitals **(d)** Randomly

43. Hund’s rule favours:

(a) No spin **(b)** Maximum unpaired electrons **(c)** Empty orbitals **(d)** Maximum pairing

44. s orbital shape is:

(a) Spherical **(b)** Dumb-bell **(c)** Four-lobed **(d)** Linear

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45. p orbital shape is:

(a) Ring **(b)** Dumb-bell **(c)** Octahedral **(d)** Spherical

46. Number of d orbitals:

(a) 7 **(b)** 5 **(c)** 3 **(d)** 1

47. Maximum electrons in a shell is:

(a) n **(b)** n^2 **(c)** 2n **(d)** $2n^2$

48. Chromium configuration is:

(a) [Ar] 3d5 4s1 **(b)** [Ar] 3d6 **(c)** [Ar] 3d4 4s2 **(d)** [Ar] 4s2 4p1

49. Copper configuration is:

(a) [Ar] 3d9 4s2 **(b)** [Ar] 3d10 4s2 **(c)** [Ar] 3d10 4s1 **(d)** [Ar] 4s2 4p1

50. A free radical has:

(a) Unpaired electron **(b)** Positive charge only **(c)** No electrons **(d)** Only paired electrons

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Answer Key with Brief Reasons

1. Answer: (b) Protons — Z identifies the element.

2. Answer: (a) Protons + neutrons — A is nucleon number.

3. Answer: (a) 14 — N = A - Z = 27 - 13.

4. Answer: (a) 10 — Al loses three electrons.

5. Answer: (b) Positive plate — Electrons are negative.

6. Answer: (b) Undeflected — They have no charge.

7. Answer: (d) Much smaller mass — Same charge but far less mass.

8. Answer: (c) New inner shell — Core electrons are much harder to remove.

9. Answer: (d) Shell size/energy — n is main energy level.

10. Answer: (a) Subshell shape — l defines s, p, d, f.

11. Answer: (b) 1 — s=0, p=1, d=2, f=3.

12. Answer: (d) 3 — 2l+1 = 3.

13. Answer: (a) 5 — 2l+1 = 5.

14. Answer: (a) 10 — Five orbitals hold two each.

15. Answer: (b) +1/2 and -1/2 — Electron spin has two values.

16. Answer: (d) No same four quantum numbers — An orbital holds opposite spins only.

17. Answer: (c) Fill lowest energy first — Electrons enter lowest available orbitals.

18. Answer: (d) Singly fill degenerate orbitals — Parallel spins minimize repulsion.

19. Answer: (a) Spherical — s is non-directional sphere.

20. Answer: (c) Dumb-bell — p has two lobes.

21. Answer: (c) 5 — d subshell has five orientations.

22. Answer: (c) 1s2 2s2 2p3 — N has seven electrons.

23. Answer: (a) Half-filled d stability — 3d5 4s1 is more stable.

24. Answer: (a) Lower energy — Aufbau order follows energy.

25. Answer: (c) Quantized energy levels — Only definite transitions occur.

26. Answer: (b) Visible region — Balmer transitions end at n=2.

27. Answer: (a) UV — Transitions end at n=1.

28. Answer: (b) Identify elements — Each element has unique lines.

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29. Answer: (a) 2 electrons — Pauli principle limits occupancy.

30. Answer: (d) 3 — l = 0, 1, 2.

31. Answer: (b) Atomic number — Z equals protons.

32. Answer: (d) Protons + neutrons — A is mass number.

33. Answer: (b) Negative plate — Protons are positive.

34. Answer: (c) Undeflected — No charge.

35. Answer: (a) Line spectra — Quantized transitions give lines.

36. Answer: (d) Quantized — Only certain energies are allowed.

37. Answer: (a) New inner shell — Core electrons are harder to remove.

38. Answer: (d) Subshell shape — l defines s, p, d, f.

39. Answer: (d) Orbital orientation — m defines orientation.

40. Answer: (a) +1/2 and -1/2 — Two spin states.

41. Answer: (b) Four quantum numbers — Unique quantum state.

42. Answer: (a) Lower energy first — Energy ordering.

43. Answer: (b) Maximum unpaired electrons — Degenerate orbitals fill singly.

44. Answer: (a) Spherical — s is non-directional.

45. Answer: (b) Dumb-bell — Two lobes.

46. Answer: (b) 5 — l=2 gives five orientations.

47. Answer: (d) $2n^2$ — Shell capacity.

48. Answer: (a) [Ar] 3d5 4s1 — Half-filled d stability.

49. Answer: (c) [Ar] 3d10 4s1 — Fully filled d stability.

50. Answer: (a) Unpaired electron — Radicals are reactive.

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